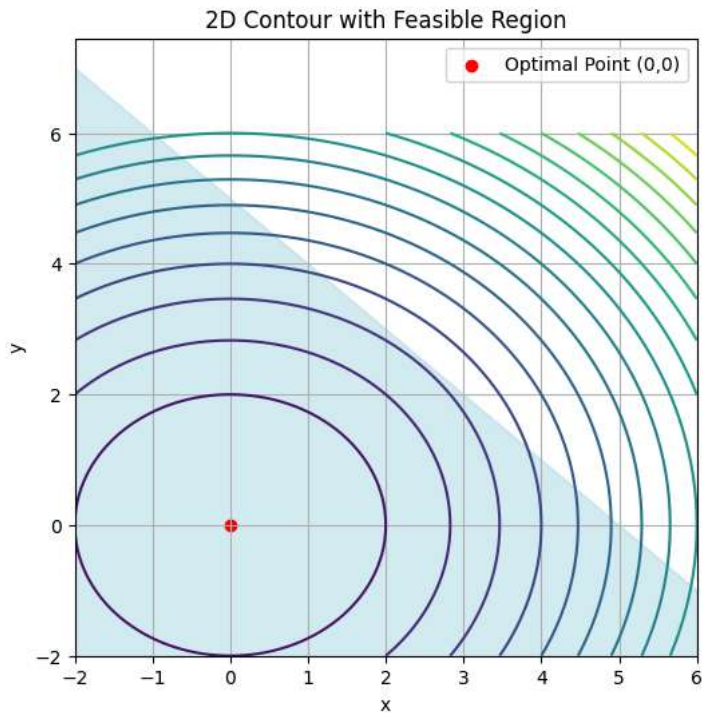
241801053 J B DHEJASVI KKT

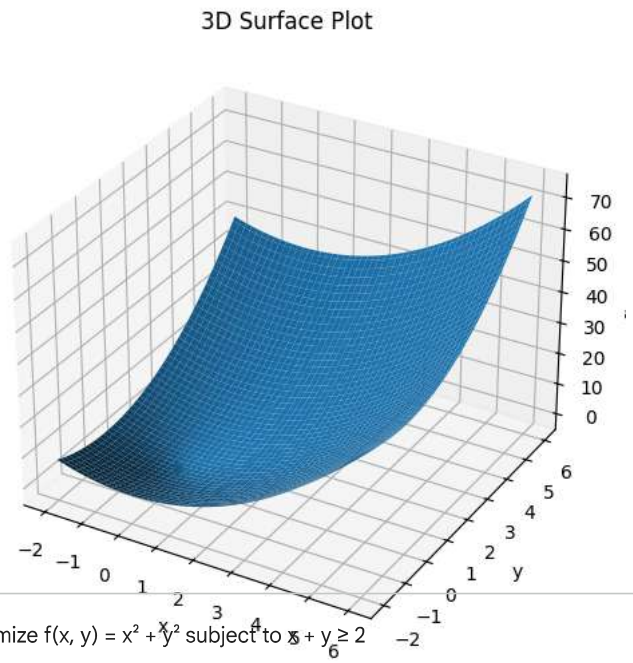
Instructions For each problem:

1. Formulate the Lagrangian and derive KKT conditions.
2. Solve analytically for optimal solution.
3. Write Python code to generate: • 2D contour plot with feasible region • 3D surface plot
4. Mark optimal point and identify active constraints.
5. Provide a brief interpretation.

Problems Q1. Minimize $f(x, y) = x^2 + y^2$ subject to $x + y \leq 5$

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Create grid
x = np.linspace(-2, 6, 100)
y = np.linspace(-2, 6, 100)
X, Y = np.meshgrid(x, y)

# Function
Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Feasible region: x + y <= 5
plt.fill_between(x, -2, 5-x, color='lightblue', alpha=0.5)

# Optimal point
plt.scatter(0, 0, color='red', label='Optimal Point (0,0)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.show()

# -------------------------------
# 3D Surface Plot
# -------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(0, 0, 0)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

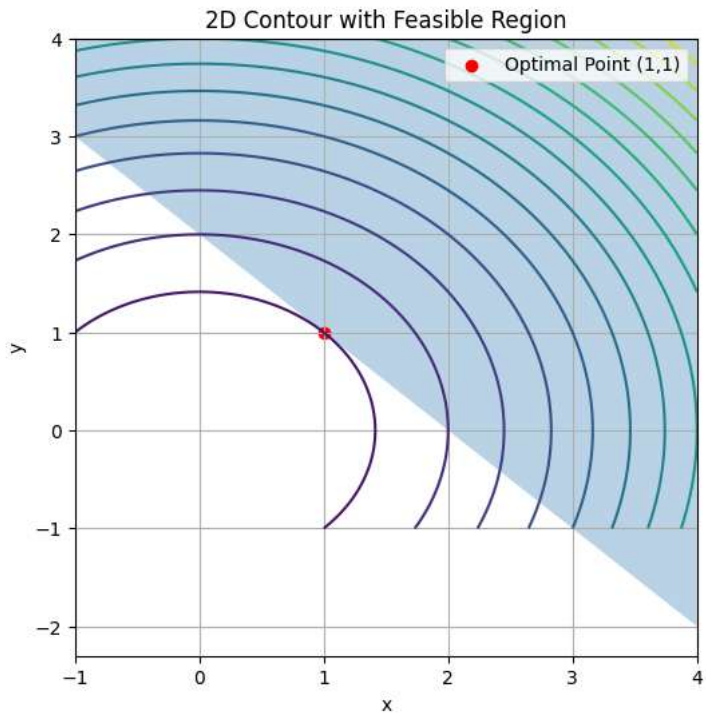


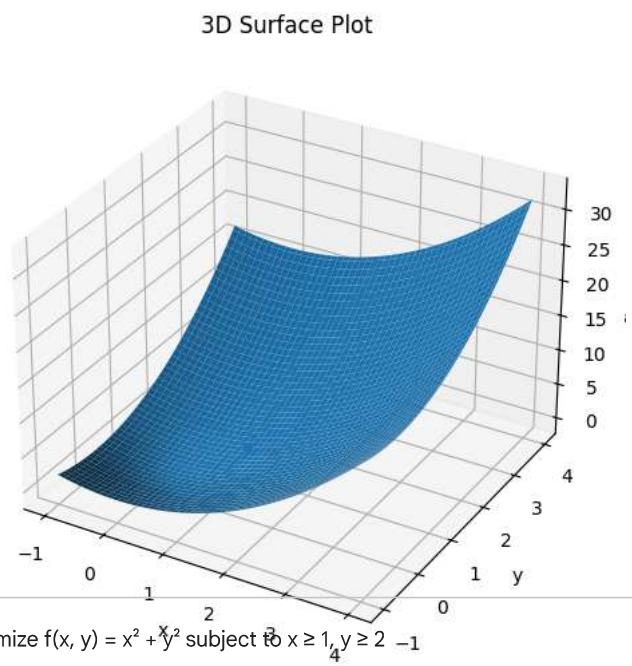


Q2. Minimize f(x, y) = x² + y² subject to x + y ≥ 2

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-1, 4, 100)
y = np.linspace(-1, 4, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Feasible region: x + y >= 2
plt.fill_between(x, 2-x, 4, alpha=0.3)
```

```
# Optimal point
plt.scatter(1, 1, color='red', label='Optimal Point (1,1)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.show()

# -------------------------------
# 3D Surface Plot
# -------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(1, 1, 2)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

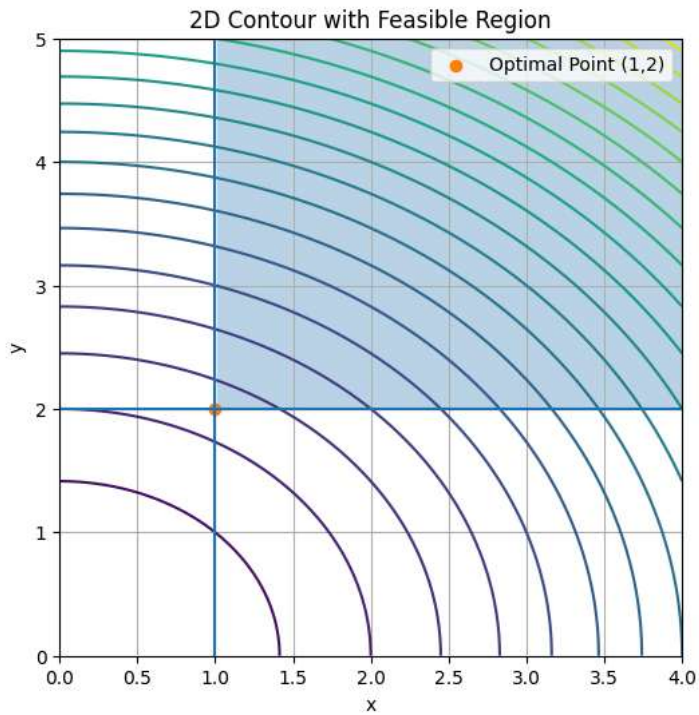


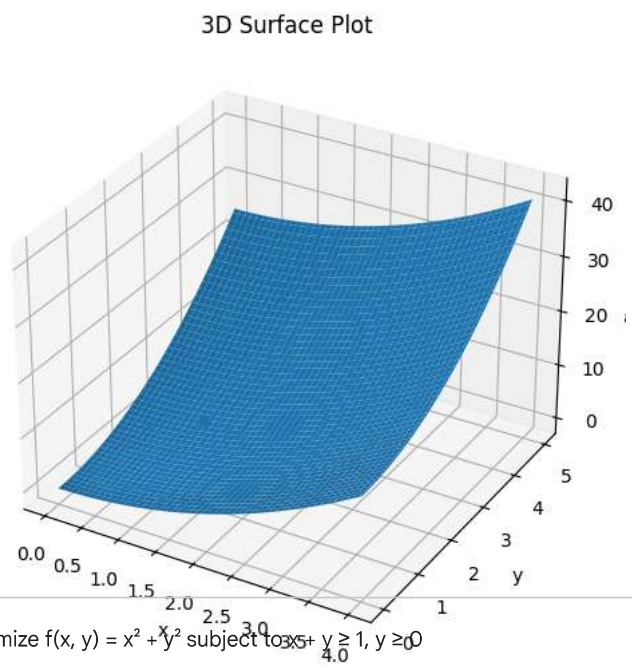


Q3. Minimize f(x, y) = $x^2 + y^2$ subject to x ≥ 1, y ≥ 2

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(0, 4, 100)
y = np.linspace(0, 5, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Feasible region: x >=1, y >=2
plt.fill_between(x, 2, 5, where=(x>=1), alpha=0.3)
```

```
# Draw boundary lines
plt.axvline(1)
plt.axhline(2)

# Optimal point
plt.scatter(1, 2, label='Optimal Point (1,2)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.show()

# ------------------------------
# 3D Surface Plot
# ------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(1, 2, 5)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

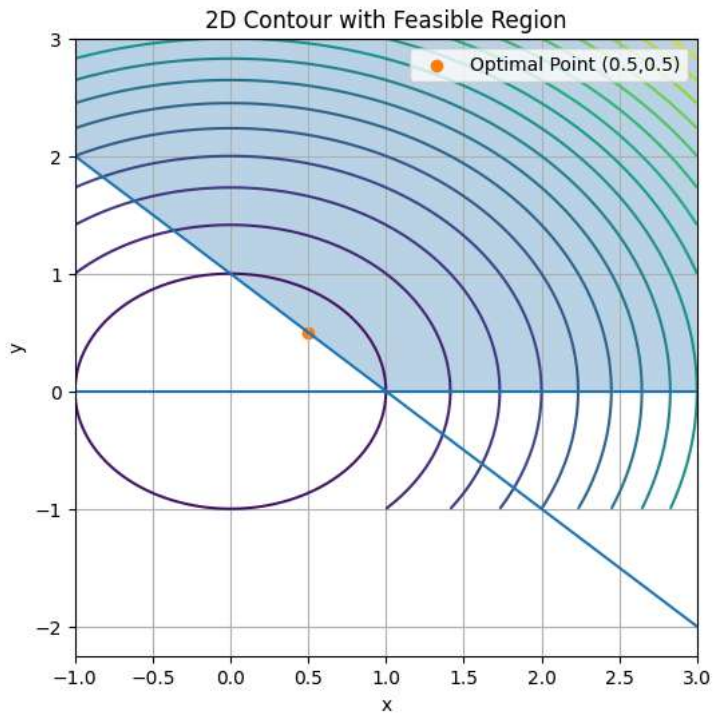


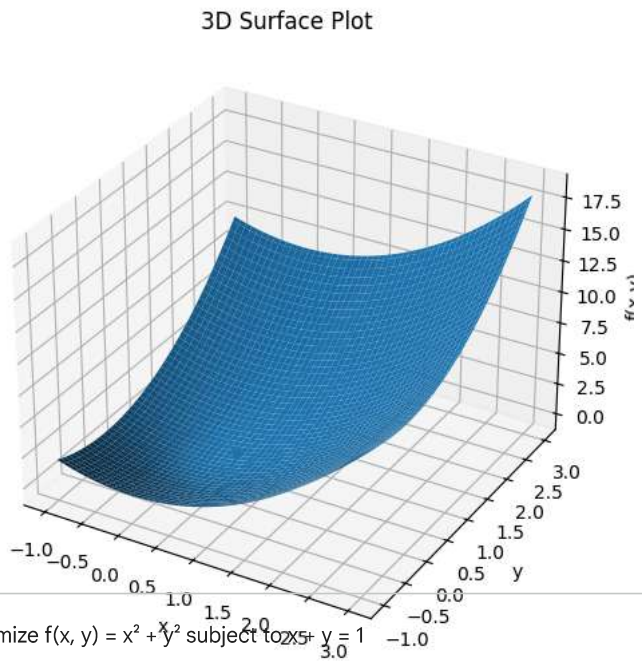


Q4. Minimize f(x, y) = x² + y² subject to x + y ≥ 1, y ≥ 0

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-1, 3, 100)
y = np.linspace(-1, 3, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Feasible region: x+y>=1 and y>=0
plt.fill_between(x, np.maximum(0, 1-x), 3, alpha=0.3)
```

```
# Boundary lines
plt.plot(x, 1-x)   # x+y=1
plt.axhline(0)     # y=0

# Optimal point
plt.scatter(0.5, 0.5, label='Optimal Point (0.5,0.5)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.show()

# ------------------------------
# 3D Surface Plot
# ------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(0.5, 0.5, 0.5)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

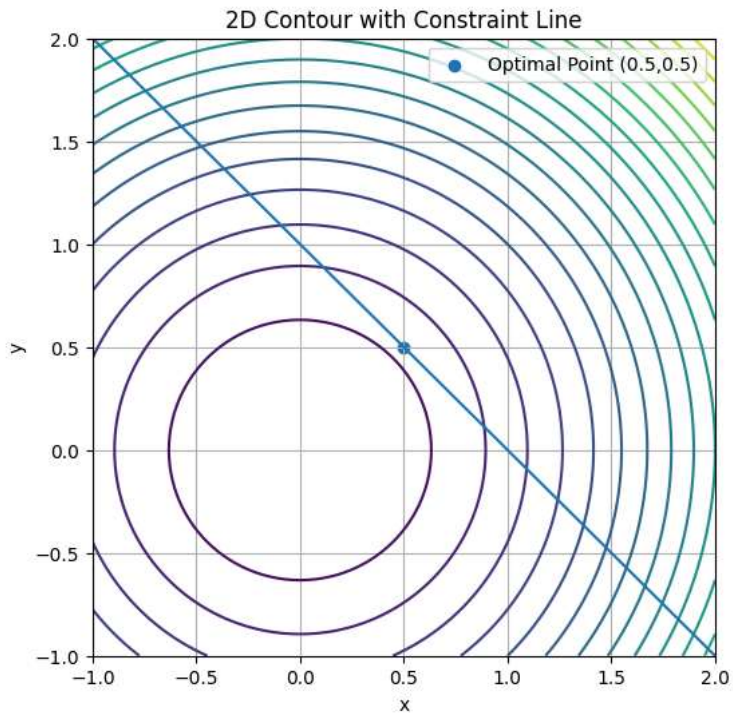


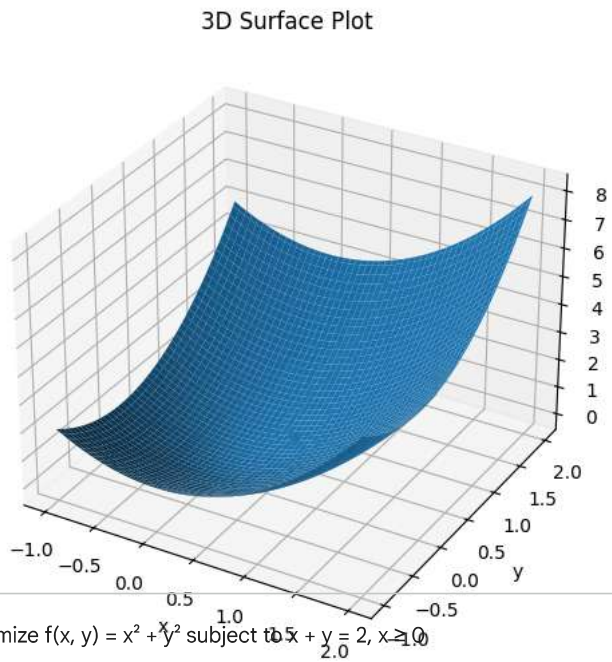


Q5. Minimize f(x, y) = x² + y² subject to x + y = 1

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-1, 2, 100)
y = np.linspace(-1, 2, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Constraint line: x + y = 1
plt.plot(x, 1-x)
```

```
# Optimal point
plt.scatter(0.5, 0.5, label='Optimal Point (0.5,0.5)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Constraint Line')
plt.legend()
plt.grid()
plt.show()

# -------------------------------
# 3D Surface Plot
# -------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(0.5, 0.5, 0.5)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

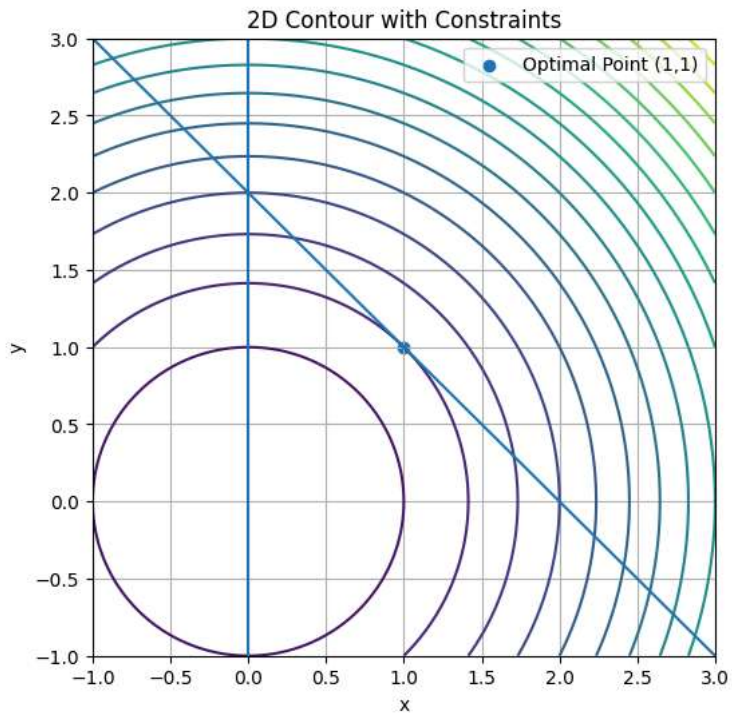


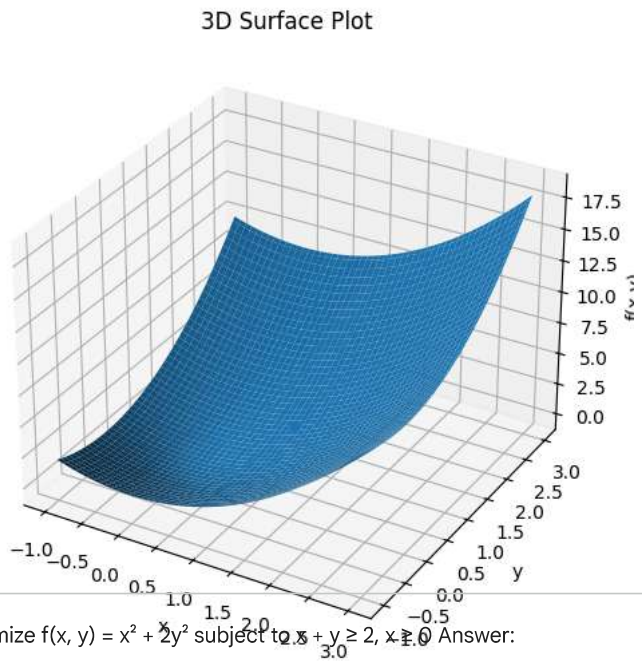


Q6. Minimize f(x, y) = x² + y² subject to x + y = 2, x ≥ 0

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-1, 3, 100)
y = np.linspace(-1, 3, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Constraint line: x + y = 2
plt.plot(x, 2-x)
```

```
# Feasible region: x >= 0
plt.axvline(0)

# Optimal point
plt.scatter(1, 1, label='Optimal Point (1,1)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Constraints')
plt.legend()
plt.grid()
plt.show()

# -------------------------------
# 3D Surface Plot
# -------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(1, 1, 2)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```

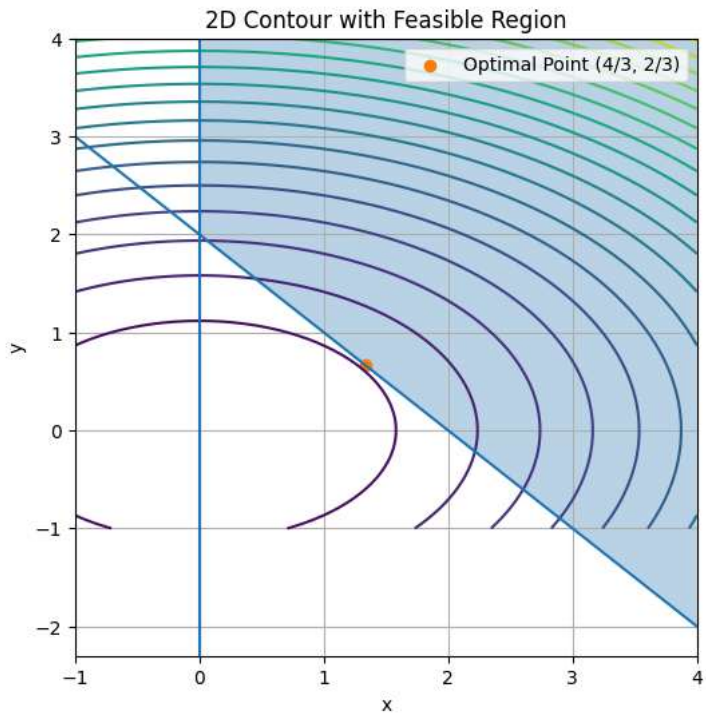


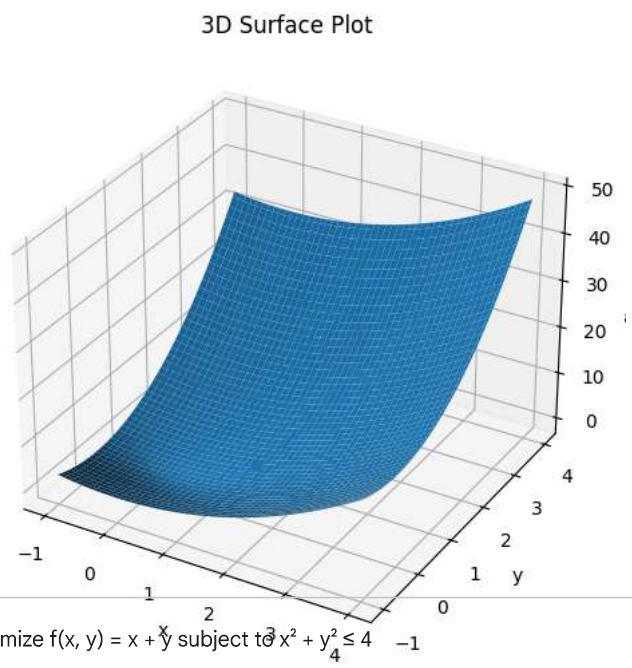


Q7. Minimize f(x, y) = $x^2 + 2y^2$ subject to x + y ≥ 2, x ≥ 0 Answer:

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-1, 4, 100)
y = np.linspace(-1, 4, 100)
X, Y = np.meshgrid(x, y)

Z = X**2 + 2*Y**2

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Feasible region: x+y>=2 and x>=0
plt.fill_between(x, 2-x, 4, where=(x>=0), alpha=0.3)
```

```
# Boundary lines
plt.plot(x, 2-x)
plt.axvline(0)

# Optimal point
plt.scatter(4/3, 2/3, label='Optimal Point (4/3, 2/3)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.show()

# ------------------------------
# 3D Surface Plot
# ------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)

# Mark optimal point
ax.scatter(4/3, 2/3, 8/3)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x,y)')
ax.set_title('3D Surface Plot')

plt.show()
```





Q8. Maximize f(x, y) = x + y subject to $x^2 + y^2 \leq 4$

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Grid
x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)

Z = X + Y

# -------------------------------
# 2D Contour Plot
# -------------------------------
plt.figure(figsize=(6,6))
plt.contour(X, Y, Z, levels=20)

# Circle boundary
theta = np.linspace(0, 2*np.pi, 100)
plt.plot(2*np.cos(theta), 2*np.sin(theta))
```

```
# Optimal point
plt.scatter(np.sqrt(2), np.sqrt(2), label='Optimal Point')

plt.xlabel('x')
plt.ylabel('y')
plt.title('2D Contour with Feasible Region')
plt.legend()
plt.grid()
plt.axis('equal')
plt.show()

# ------------------------------
# 3D Surface Plot
# ------------------------------
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(X, Y, Z)
```